MICROECONOMICS

MICRO-13

Time: 5 minutes

Outcome: Identify Efficient Quantity & Allocation.

Difficulty: ●●○ Intermediate

Efficient Quantity & Allocation

THE CORE IDEA

The efficient quantity includes every unit whose buyer value is at least as large as its opportunity cost. Efficiency pairs the highest-value buyers with the lowest-cost sellers and continues exchange while the next buyer's willingness to pay covers the next seller's cost. Producing beyond that point destroys surplus, while stopping before it leaves gains unrealized.

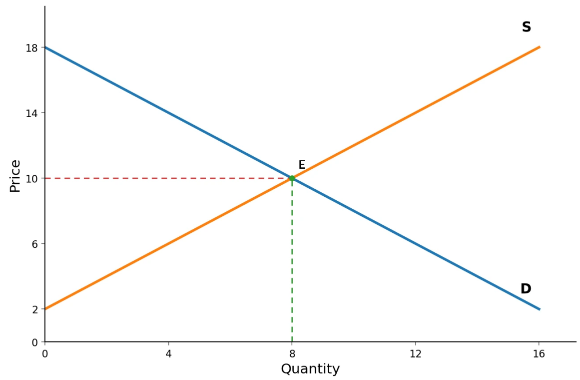
HOW TO RECOGNIZE IT

- Order potential trades from the highest buyer value and the lowest seller cost.
- Complete every trade for which willingness to pay is at least as large as opportunity cost.
- Stop when the next seller's cost exceeds the next buyer's value.
- An efficient allocation sends the good to the highest-value buyers and production to the lowest-cost sellers.

WATCH OUT

A common mistake is to equate more output with more efficiency. The forced extra unit contributes $23 - $29 = -$6 to total surplus.

WORKED EXAMPLE: FINDING THE EFFICIENT QUANTITY



Demand and supply intersect at 8 units and $10. At that quantity, the marginal buyer's value equals the marginal seller's cost. Units below 8 create positive gains from trade; units above 8 cost more than buyers value them. The efficient allocation therefore produces 8 units.

CHECK YOURSELF

Five buyers each want one studio pass, with willingness-to-pay values [27, 23, 18, 13, 8]. Five sellers can each supply one, with costs [4, 10, 15, 21, 25]. Ignore any announced market price. Rank buyers from highest to lowest value and sellers from lowest to highest cost. Which statement correctly identifies the efficient cutoff?

READY? Return to the game and master this concept.